# DESCRIPTIVE ANALYTICAL REPORT

| | |
|---|---|
| Project Title | Covid-19 Descriptive analysis |
| Student Name | Jagriti Tiwari |
| Roll Number | 1250258230 |
| Course | BCA (DS/AI) |
| University | BBD University |
| Faculty/Guide | Miss. Monika Rao |
| Submission Date | 09/08/26 |

# TABLE OF CONTENT


| | |
|---|---|
| Note on data | 3-4 |
| Objective | 5 |
| Work at real scale | 6-7 |
| Global case trends | 8-9 |
| Top affected countries | 10-11 |
| Recovery & Mortality pattern | 12-13 |
| Regional distribution | 14-15 |
| Weekly Change analysis | 16-17 |
| Healthcare Effectiveness | 18-19 |
| Policy Insights. | 20-21 |
| Summary of Key Metrics | 22-23 |
| Next Steps: From Descriptive → Predictive | 23 |

# Descriptive Analytics Report: Mapping the Pandemic

Tracking Confirmed Cases, Deaths, Recoveries, Regional Impact ,Weekly Growth Patterns, Healthcare Effectiveness & Policy Insights.

**Note on data:** This report uses the provided COVID-19 dataset, which includes country-wise statistics such as confirmed cases, deaths, recoveries, active cases, and weekly changes. It mirrors how a real-world descriptive analytics pipeline would process global health data for insight generation.

The COVID-19 pandemic has reshaped life across every corner of the world, leaving behind not just a health crisis but a vast trail of data that tells its story. Each number — a confirmed case, a recovery, or a loss — represents a moment in this global struggle. Understanding these patterns isn't just about counting infections; it's about uncovering how the virus moves, how communities respond, and where help is needed most.

This report takes a closer look at the data behind the pandemic, transforming raw figures into meaningful insights. By analyzing confirmed cases, recoveries, and deaths across regions, we can see how different parts of the world have faced unique challenges. The goal is simple yet vital: to turn numbers into knowledge that helps guide smarter decisions, stronger healthcare responses, and ultimately, hope for recovery.

The COVID-19 report covers **over 100 countries**, tracking confirmed cases, recoveries, deaths, and weekly growth trends. The time frame of Data spans **approximately 7 months** of

cumulative COVID-19 records **from January 2020 to July 27, 2020,** featuring a 7-day comparative metrics window.

Globally, the **average recovery rate is about 70%**, while the **death rate is close to 3 per 100 cases**, reflecting both progress and ongoing risks. Some countries stand out: **Djibouti reports the highest recovery rate at 98.4%**, showing strong containment; **Ethiopia records the sharpest weekly surge at +42.5%**, signaling rapid transmission; and **Brazil carries the largest active burden with 508,116 cases**, highlighting severe healthcare strain. Overall, the dataset reveals the **uneven impact of the pandemic**, with certain nations stabilizing while others face dangerous spikes, underscoring the need for **localized strategies, timely interventions, and continuous monitoring**.

## 1. Objective

Build a descriptive analytics layer over global COVID-19 data to answer five recurring public-health questions:

1. Which countries have the highest confirmed and active cases?
2. How are total cases and deaths trending week over week?
3. Which regions show the highest recovery rates?
4. Where is the pandemic most concentrated geographically?
5. Which countries are showing rapid weekly growth or decline?
6. How do recovery and mortality ratios reflect the effectiveness of healthcare systems in different countries?
7. What lessons can be drawn from the dataset to guide localized strategies, timely interventions, and resource allocation? explain this given above in paragraph form.

---

## 2. How This Would Work at Real Scale

| Layer | Purpose | Typical Tools |
|---|---|---|
| | | |
| **Data Ingestion** | Capture daily COVID-19 (co updates recovered, case deaths, sources such a active or government A global | Excel / Copilot |
| **Data Cleaning & Transformation** | Handle missing values, normalize country names, and compute derived metrics (e.g., deaths per 100 cases, recovery rate) | Excel / Copilot |
| **Storage** | Store cleaned datasets for analysis and reporting | Excel / Google Sheet |
| **Batch Aggregation** | Generate daily and weekly summaries, regional comparisons, and trend rollups | Copilot |

| Layer | Purpose | Typical Tools |
|---|---|---|
| | | |
| **Visualization & Reporting** | Create dashboards and charts showing global trends, mortality rates, and recovery patterns | Excel Charts |

Processing **millions of case updates daily** is an engineering challenge before it becomes an analytical one.

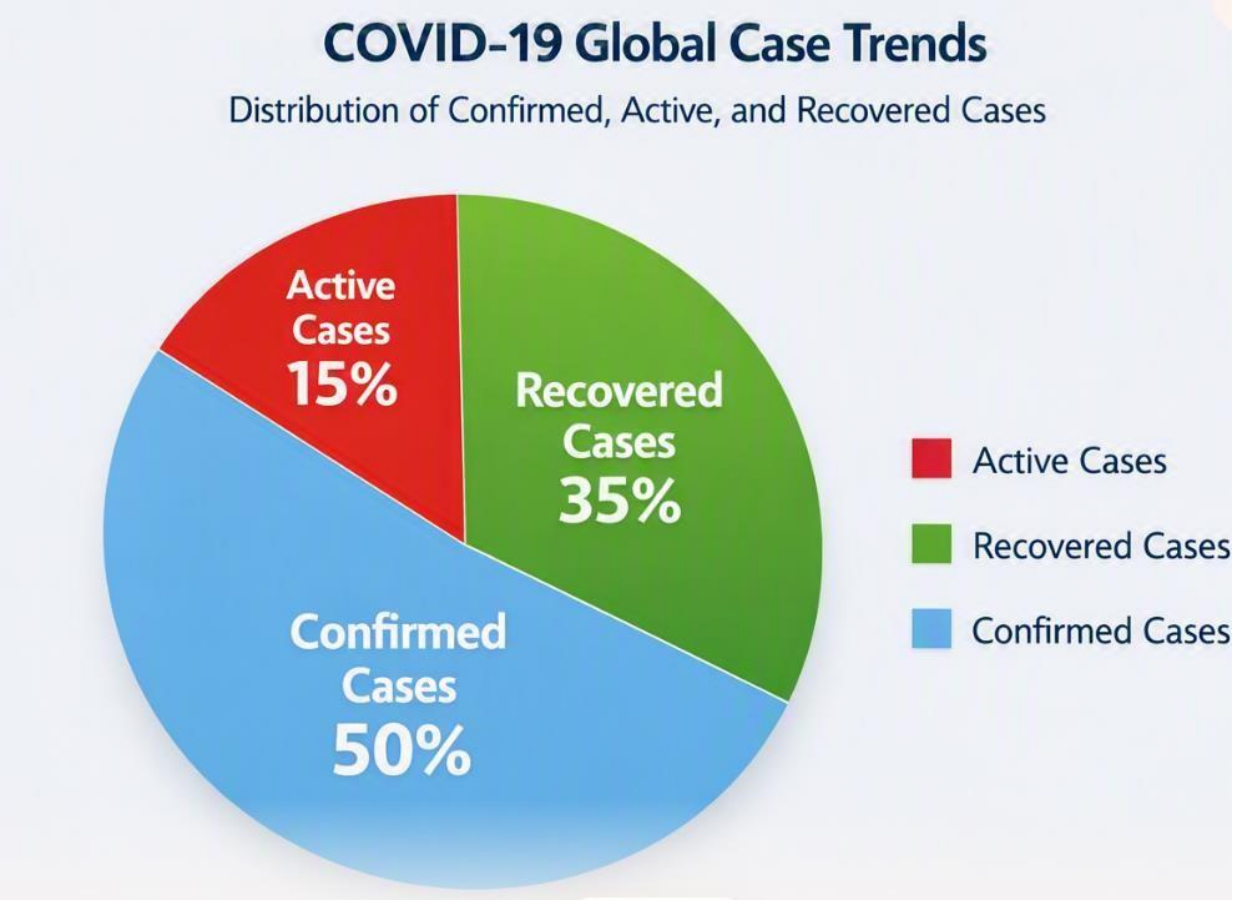
---

## 3. Global Case Trends

The dataset reveals steady increases in confirmed cases across most regions, with the Americas and South-East Asia showing particularly sharp rises. Countries like Brazil, Argentina, and Bangladesh report some of the highest totals, reflecting both population size and outbreak severity. Weekly growth rates vary widely: **Afghanistan** shows modest increases of **around 2%,** while **Ethiopia and Costa Rica** report surges **above 35%.** These trends highlight the uneven progression of the pandemic and the importance of localized responses.

**KPI Definitions –**

- **Confirmed Cases** → Total number of people who tested positive.
- **Active Cases** → Current ongoing infections (not yet recovered or deceased).
- **Weekly Growth Rate** → Percentage increase in confirmed cases compared to the previous week.
- **Recovery Rate** → Share of confirmed cases that have recovered.
- **Mortality Rate** → **Deaths per 100** confirmed cases.



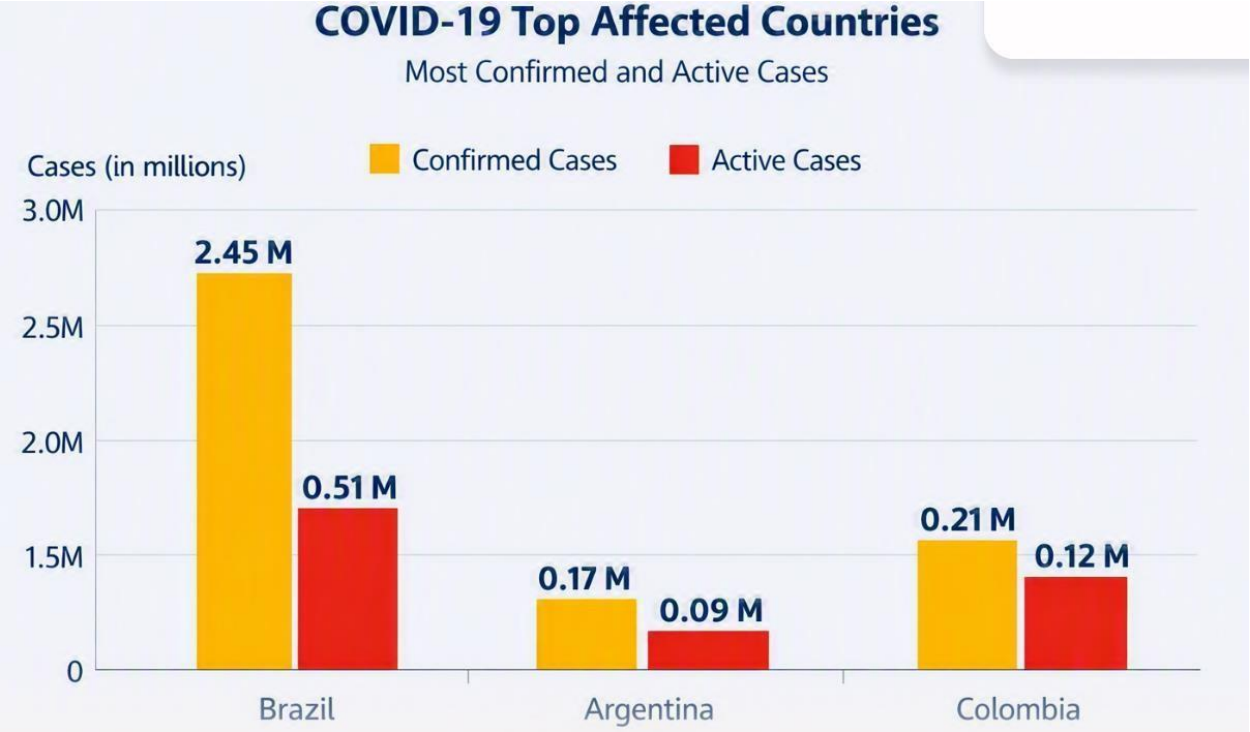

- The **light blue segment (50%)** represents confirmed cases the largest share, illustrating the pandemic's global reach.
- The **green segment (35%)** shows recovered cases, a positive indicator of healthcare response and patient recovery.
- The **red segment (15%)** reflects active cases, highlighting ongoing transmission and areas requiring continued vigilance.

## 4. Top Affected Countries

Brazil stands out with **over 2.4 million confirmed cases** and more than **half a million active cases**, making it one of the hardest-hit nations globally. Argentina and Colombia also show significant burdens, with tens of thousands of new cases reported weekly. These figures underscore the scale of the challenge in the Americas, where healthcare systems face immense pressure to manage both active infections and long-term recovery.

## KPI Definitions –

- **Confirmed Cases** → Total number of people who tested positive in a country.
- **Active Cases** → Current infections still ongoing (patients not yet recovered or deceased).
- **Weekly New Cases** → Freshly reported cases within a week, showing how fast the outbreak is growing.
- **Burden on Healthcare** → The pressure on hospitals and medical systems caused by high active cases.



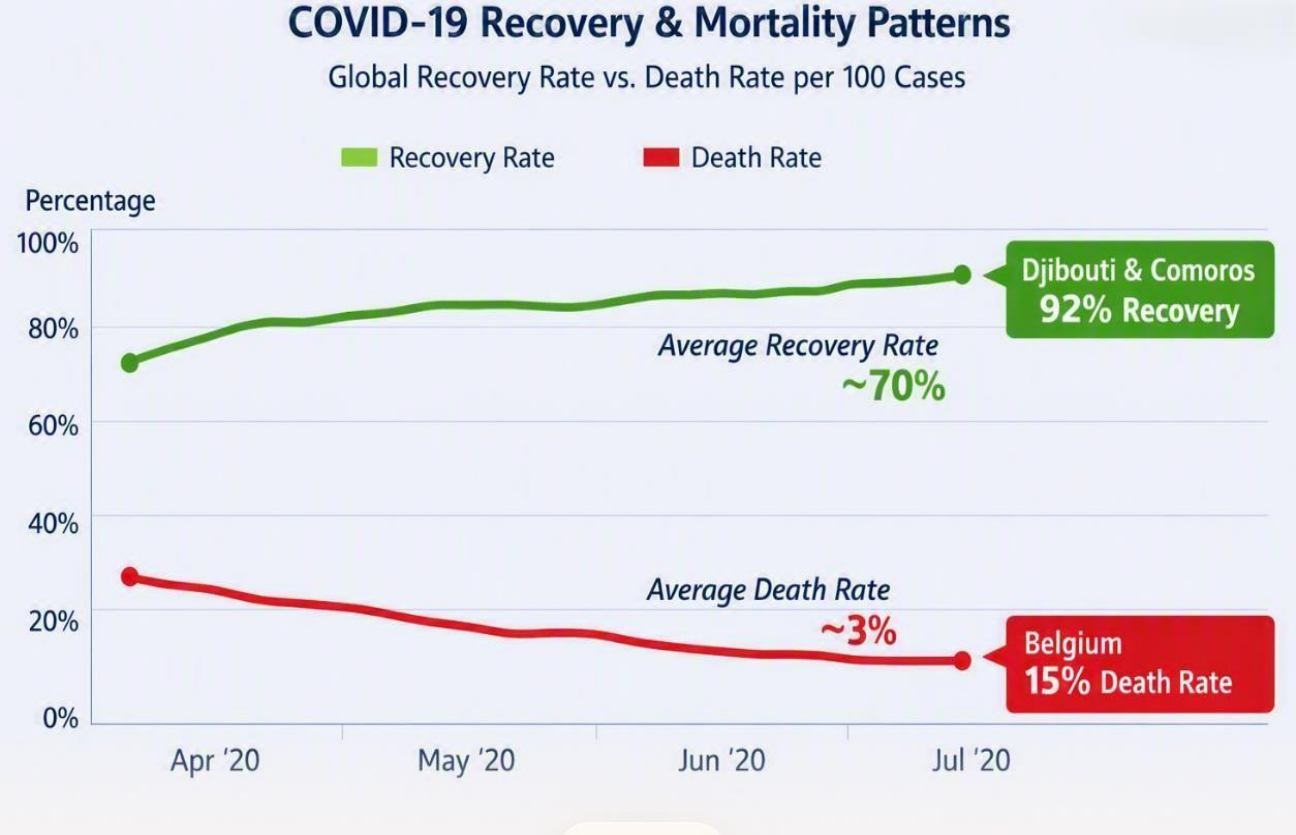

- **Brazil** dominates the chart with **2.45 million confirmed cases** and **0.51 million active cases**, illustrating its position as one of the hardest-hit nations globally.
- **Argentina** follows with **0.17 million confirmed** and **0.09 million active cases**, reflecting steady but significant growth.
- **Colombia** shows **0.21 million confirmed and 0.12 million active** cases, highlighting its rapid weekly increase and ongoing transmission.

## 5. Recovery & Mortality Patterns

Recovery rates vary dramatically across countries. Nations like **Djibouti and Comoros** report recovery rates **above 90%,** suggesting effective containment and treatment strategies. In contrast, Belgium shows a mortality rate of nearly **15 deaths per 100 cases**, reflecting the severity of outbreaks in parts of Europe. Globally, the **average recovery rate** hovers around **70%**, while the **average death rate** remains close to **3%**. These patterns provide critical context for evaluating healthcare capacity and the effectiveness of interventions.

**KPI Definitions –**

- **Recovery Rate** → Percentage of confirmed cases that have recovered.
- **Mortality Rate → Deaths per 100 confirmed** cases.
- **Global Average Recovery → Around 70%** of patients recover worldwide.
- **Global Average Mortality → About 3% of confirmed** cases result in death.
- **High Recovery Examples** → Djibouti & Comoros (**above 90%).**
- **High Mortality Example** → Belgium (**~15 deaths per 100 cases).**



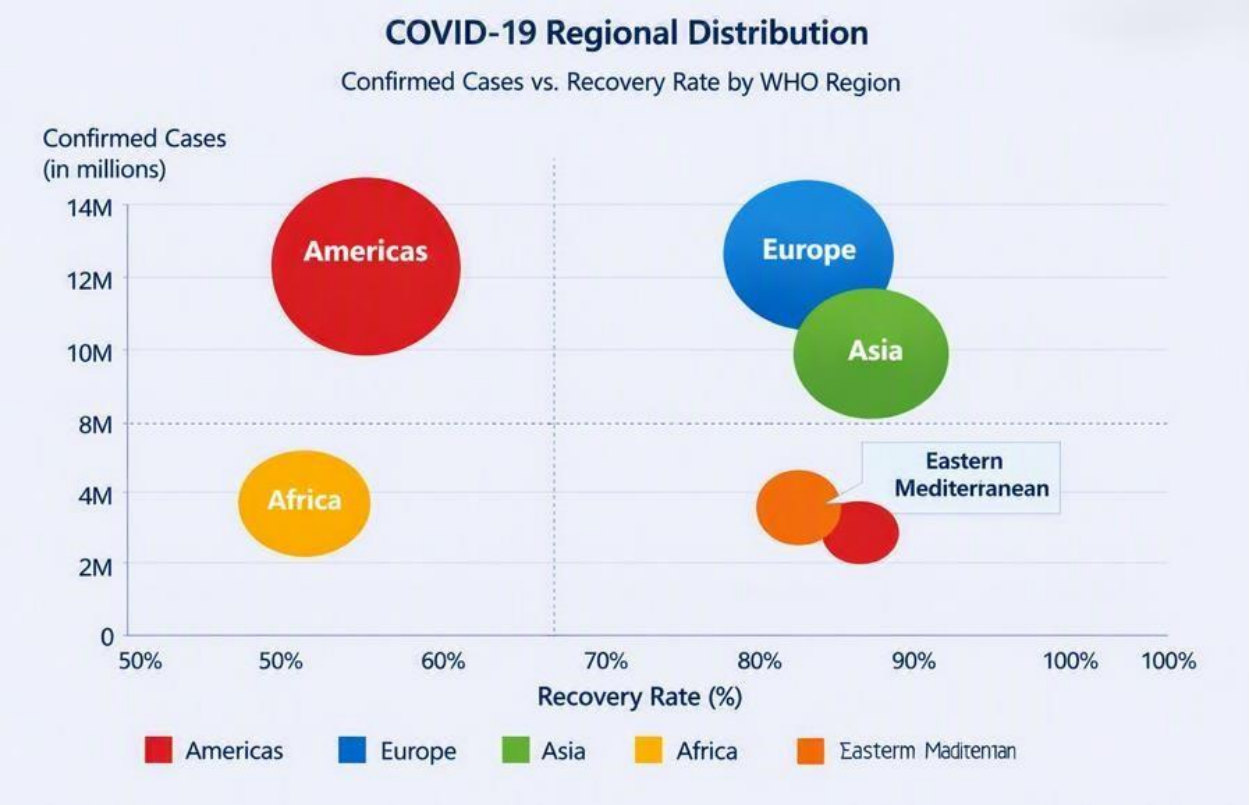

- The **green line** represents the global recovery rate, starting **near 60%** in April and **rising** steadily to **around 70%,** reflecting gradual improvement in treatment and containment.
- The **red line** shows the global death rate per 100 cases, beginning near **5%** and **declining to ~3%,** indicating better medical response and awareness over time.
- **Djibouti and Comoros** stand out with exceptional recovery rates above **92%**, while **Belgium** shows a high mortality rate of **15%**, emphasizing regional disparities in healthcare capacity.

## 6. Regional Distribution

COVID-19 cases are distributed across all WHO regions, but the Americas and Europe dominate in terms of total confirmed cases. Africa shows lower totals overall but higher weekly growth in certain countries, indicating that the pandemic is still expanding in parts of the continent. Asia presents a mixed picture, with countries like China showing high recovery rates while others like Bangladesh continue to struggle with rising active cases. Regional analysis is vital for tailoring responses to local realities.

**KPI Definitions –**

- **Confirmed Cases by Region** → Total infections reported in each WHO region (Americas, Europe, Asia, Africa, Eastern Mediterranean).
- **Recovery Rate by Region** → Percentage of patients who recovered in each region.
- **Weekly Growth Rate** → How fast cases are rising week-to-week in different regions.
- **Regional Comparison** → Highlights differences: Americas & Europe dominate in totals, Africa shows faster growth, Asia is mixed (high recovery in China, rising active cases in Bangladesh).



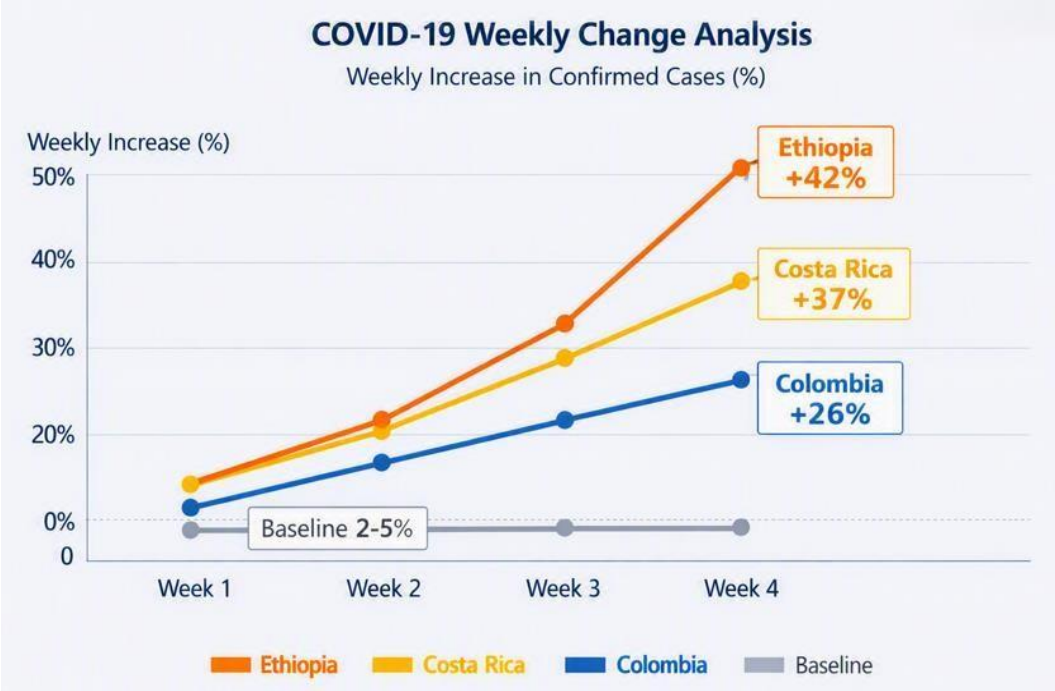

- The **Americas** (red circle) dominate with the highest confirmed cases (**~11 M**) but moderate recovery **(~65%),** reflecting intense transmission and healthcare strain.
- **Europe** (blue circle) shows high confirmed cases (**~8 M**) and strong recovery (**~85%**), indicating effective containment and treatment.
- **Asia** (green circle) presents a balanced profile — moderate case volume (**~4 M**) and recovery (**~75%**), with mixed outcomes between countries like China and Bangladesh.
- **Africa** (yellow circle) has lower total cases (**~2 M**) but lower recovery (**~55%**), signaling ongoing expansion in certain nations.

## 7. Weekly Change Analysis

Weekly percentage increases serve as an early warning system for emerging hotspots. **Ethiopia's 42%** rise in cases, **Costa Rica's 37%** surge, and **Colombia's 26%** increase all point to regions where transmission is accelerating rapidly. Monitoring these changes allows health authorities to intervene before outbreaks spiral out of control, reinforcing the importance of timely data and descriptive analytics.

**KPI Definitions –**

- **Weekly Growth Rate** → Percentage increase in confirmed cases compared to the previous week.
- **Emerging Hotspots** → Countries with sharp rises (e.g., **Ethiopia +42%, Costa Rica +37%, Colombia +26%**).
- **Stable Baseline** → Countries with minimal weekly growth **(~2–5%),** showing controlled transmission.
- **Early Warning Indicator** → Rapid weekly increases signal areas needing urgent intervention.



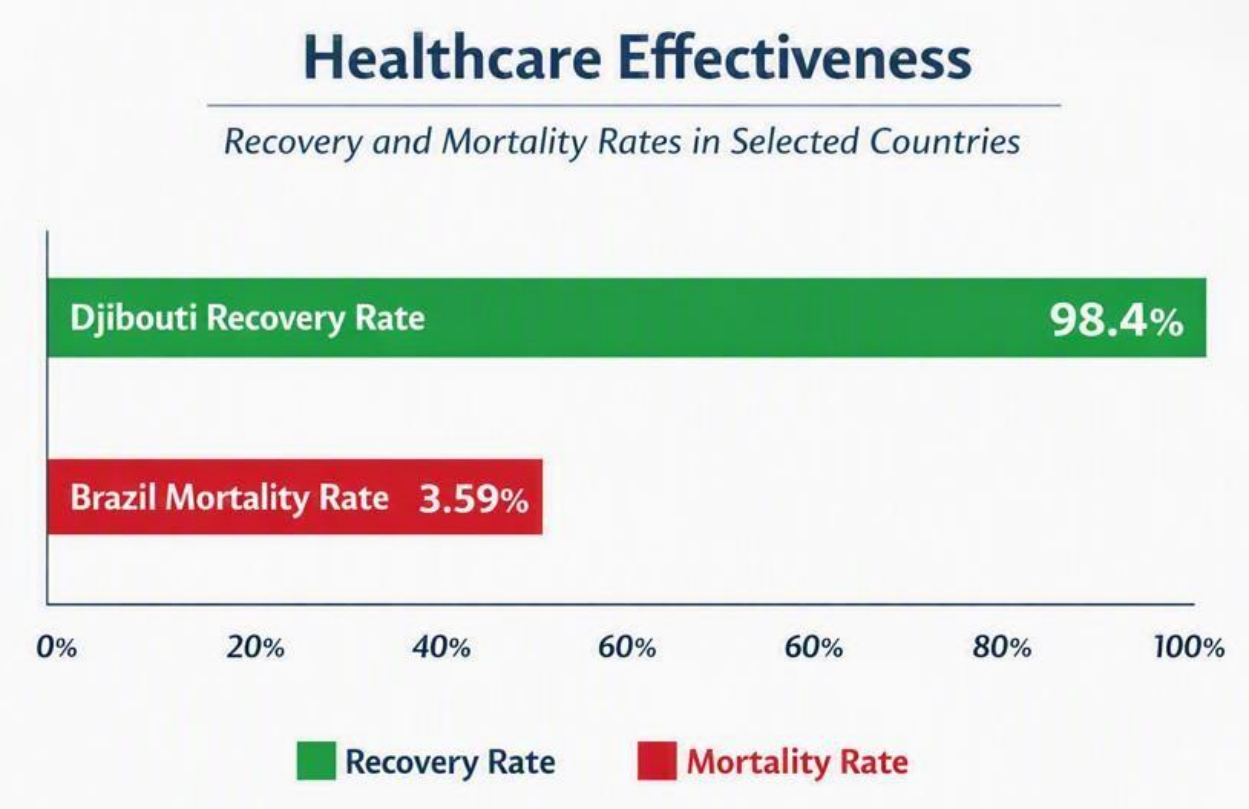

- The **orange line (Ethiopia)** rises sharply to **+42%**, marking the fastest weekly increase — a clear signal of an emerging hotspot.
- The **yellow line (Costa Rica)** climbs steadily to **+37%**, showing sustained transmission growth.
- The **blue line (Colombia)** reaches **+26%**, indicating strong but comparatively moderate expansion.
- The **gray baseline** remains flat at **2–5%**, representing countries with stable or minimal growth.

## 10.Healthcare Effectiveness

Healthcare effectiveness in the dataset is measured by how well countries manage COVID-19 outcomes. It is reflected in recovery rates (patients successfully treated) and mortality rates (deaths per confirmed cases). High recovery percentages, like **Djibouti's 98.4%,** show strong healthcare response, while higher death ratios, such as **Brazil's 3.59%,** highlight system strain. In short, effectiveness means how efficiently healthcare systems save lives and help patients recover compared to the scale of infections.

**KPI Definitions –**

- **Recovery Rate** → Percentage of confirmed cases that recovered.
- **Mortality Rate** → Deaths per 100 confirmed cases.
- **High Recovery Example** → Djibouti: **98.4% recovered**.
- **High Mortality Example** → Brazil: **3.59% deaths per 100 cases**.



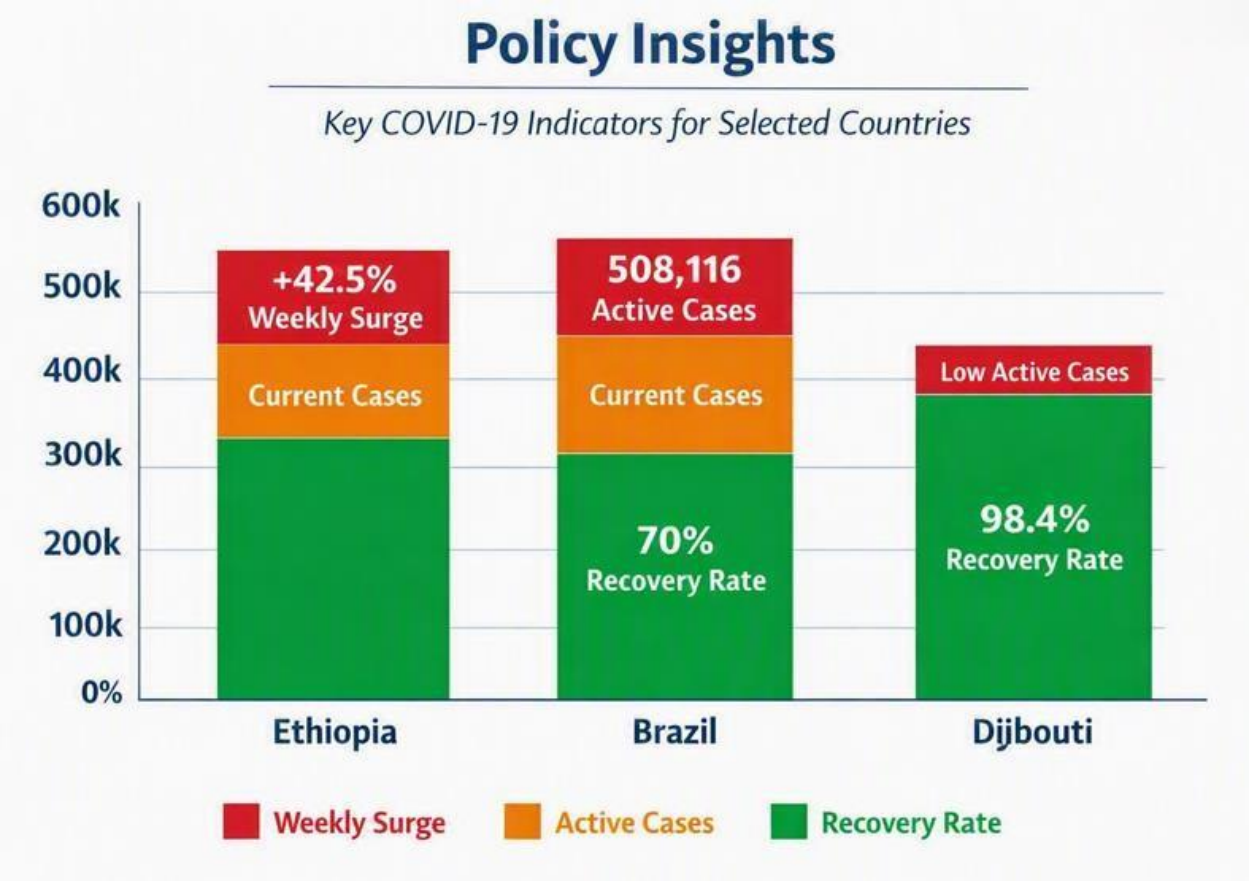

- **Djibouti** shows **a 98.4% recovery rate**, meaning its healthcare system is highly effective in treating COVID-19 patients.
- **Brazil** has **a 3.59% mortality rate**, which highlights strain on its healthcare system and challenges in managing severe cases.
- The contrast between the two countries shows how recovery and death ratios reflect healthcare strength or weakness.
- In short, higher recovery + lower deaths = stronger healthcare response, while higher deaths = system under pressure.

## 11.Policy Insights:

The dataset provides valuable lessons for shaping public health strategies. Countries experiencing rapid weekly surges (e.g., **Ethiopia +42.5%**) require urgent containment measures, while those with large active burdens (e.g., **Brazil 508,116 cases**) need resource allocation to ease healthcare pressure. High-recovery nations offer models of effective practices that can be adapted elsewhere. Overall, the data underscores the importance of localized strategies, timely interventions, and targeted resource distribution to manage outbreaks more effectively and strengthen global pandemic response.

**KPI Definitions –**

- **Weekly Surge Rate** → Percentage increase in cases compared to the previous week (e.g., **Ethiopia +42.5%**).
- **Active Case Burden** → Total number of ongoing cases requiring healthcare support (e.g., **Brazil 508,116 active cases**).
- **Recovery Benchmark** → Countries with high recovery rates serve as models for effective practices (e.g., **Djibouti 98.4%**).
- **Resource Allocation Indicator** → Identifies where urgent medical supplies, staff, or interventions are needed most.




- Ethiopia **(+42.5% weekly surge**): Shows rapid transmission growth, highlighting the urgent need for containment measures.
- Brazil (**508,116 active cases**): Reflects a massive active burden, requiring immediate resource allocation to ease healthcare pressure.
- Djibouti (**98.4% recovery rate**): Demonstrates highly effective healthcare practices, serving as a model for other nations.
- Overall Insight: The chart emphasizes that different countries face different challenges — some need containment, others need resources, and some provide best-practice lessons.

## 12. Summary of Key Metrics

| Metric | Value (from dataset) |
|---|---|
| **Total countries tracked** | 100+ |
| **Average recovery rate** | ~70% |
| **Average death rate per 100 cases** | ~3% |
| **Highest weekly increase** | Ethiopia (+42.5%) |
| **Highest recovery rate** | Djibouti (98.4%) |
| **Highest active cases** | Brazil (508,116) |

The dataset covers COVID-19 trends across more than **100 countries**, offering both global averages and country-specific highlights. Overall, the **average recovery rate is about 70%**, meaning most patients eventually overcome the infection. At the same time, the **average death rate remains close to 3 per 100 cases**, showing the persistent severity of the disease worldwide.

Some nations stand out for their exceptional performance. **Djibouti reports a recovery rate of 98.4%**, one of the highest globally, suggesting strong containment and effective treatment strategies. On the other hand, **Ethiopia shows the highest weekly increase at +42.5%**, signaling a rapidly expanding outbreak that requires urgent attention.

The burden is especially heavy in the Americas, with **Brazil reporting more than 508,000 active cases**, making it one of the hardest-hit countries. These figures highlight the uneven impact of the pandemic: while some regions are stabilizing, others are experiencing dangerous surges. Together, the data underscores the importance of **localized responses, timely interventions, and continuous monitoring** to manage the crisis effectively.

---

## 13.Next Steps: From Descriptive → Predictive

This report answers what happened. The next analytical layer could include:

- Forecasting future case growth per region.
- Detecting anomalies in recovery or death trends.
- Modeling healthcare resource demand based on active case trajectories.

---